# What each process does to the RT distribution

14 processes of the extended Wong-Wang model, each moved alone from three fitted starting points; RT distributions of 10,000 simulated trials per setting (correct responses).

Change from the lowest to the highest setting, from the lexical-decision starting point (A2):

| Process | Range (lowest → highest) | 10th | Median | 99th | 99th/med | Acc | tau |
|---|---|---|---|---|---|---|---|
| skill | x2.0 → x0.2 | +20% | -2% | -78% | -78% | -0.9 | -317 |
| practice | -2 → +4 | -2% | -0% | +5% | +5% | +1.4 | +3 |
| common drive | x0.8 → x1.3 | -31% | -34% | -39% | -8% | -9.7 | -65 |
| evidence | +0 Hz → +8 Hz | -8% | -11% | -26% | -17% | +14.3 | -45 |
| E/I excitatory | S 0.0 → S 1.0 | -29% | -33% | -51% | -26% | -10.0 | -83 |
| E/I inhibitory | S 0.0 → S 0.6 | +99% | +189% | +773% | +202% | +13.8 | +1221 |
| circuit threshold | 12 Hz → 30 Hz | +81% | +88% | +134% | +24% | +15.1 | +147 |
| instance threshold | x0.5 → x3.0 | +21% | +7% | +7% | -0% | -3.0 | +12 |
| urgency | -0.01 nA/s → 0.06 nA/s | -24% | -33% | -64% | -45% | -12.5 | -170 |
| lapse | 0.0 → 0.15 | +2% | +2% | +90% | +85% | -5.2 | +93 |
| non-decision time | -60 ms → +120 ms | +53% | +40% | +20% | -15% | +0.0 | -0 |
| retrieval accuracy | 0.5 → 1.0 | -1% | -0% | +1% | +1% | +0.3 | +3 |
| circuit strength | x0.9 → x1.18 | -31% | -42% | -79% | -64% | -10.2 | -291 |
| retrieval strength | x0.5 → x4.4 | -44% | -17% | -6% | +14% | +5.7 | +18 |

1. Columns: percent change in the 10th percentile, median and 99th percentile of correct RTs; change in the ratio of the 99th percentile to the median (tail relative to the body); accuracy (points); ex-Gaussian tau (ms). A2 = lexical decision under speed instruction, calibrated to its data. Every process from every starting point: 4_output/mechanism_fingerprints.csv.

2. Processes that act through the instance route (practice, instance threshold, retrieval accuracy and strength) do little where that route seldom wins (A2: 1%; flanker P9: 0%) and much where it often wins (StarCraft 2 Bronze S1: 59%) (Figure 2).

3. StarCraft 2 has no correct/error label, so its starting point has no evidence term and chooses at chance; its accuracy values are not interpretable.

4. Sources: 4_output/atlas/sweeps.json (3_analyses/18_mechanism_atlas.py; seed 20261107).

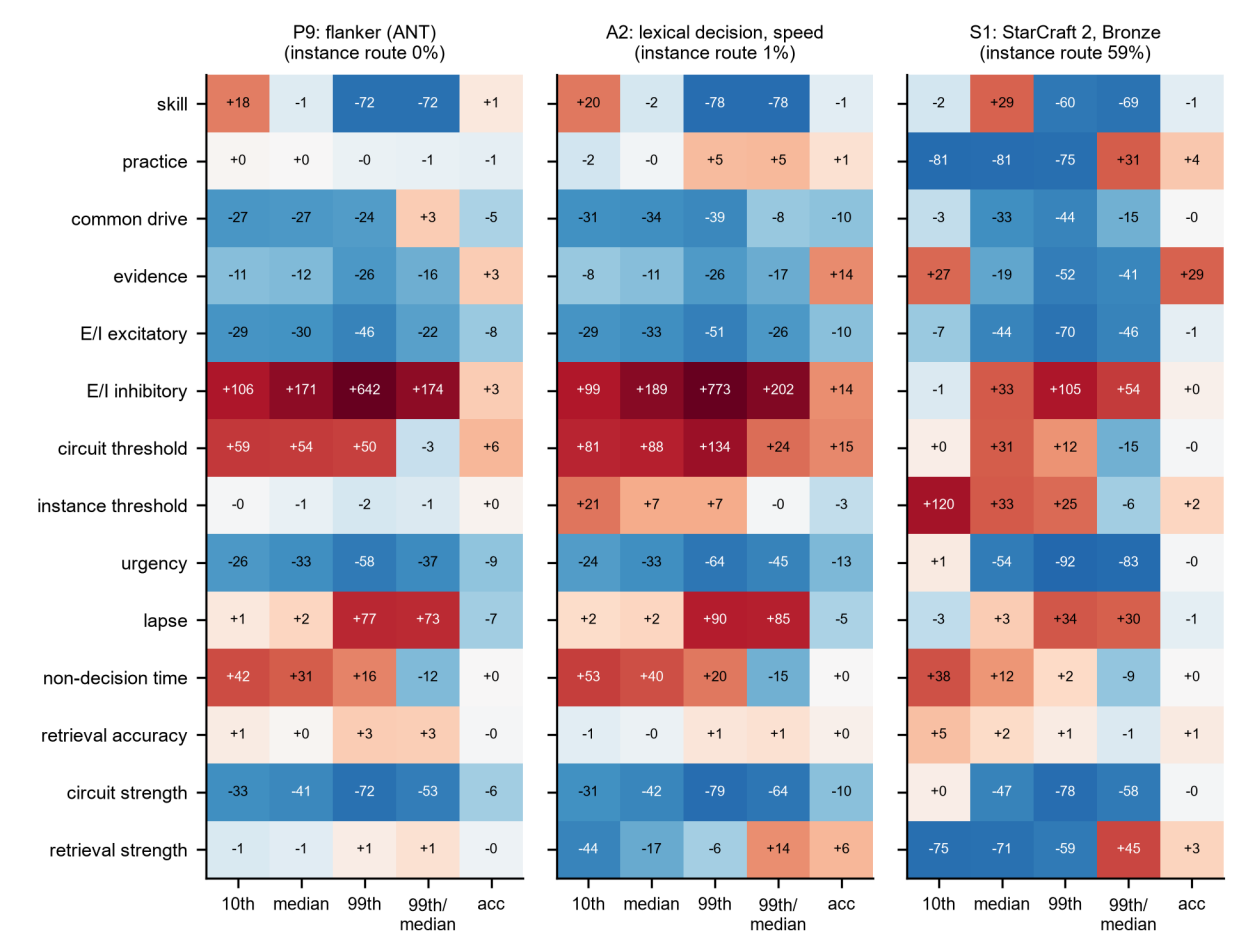



Figure 1. Fingerprint of each process: change from its lowest to its highest setting, from each starting point. Percent change at the 10th percentile, median and 99th percentile, change in the 99th/median ratio (tail relative to the body), and accuracy in points. Red = increase, blue = decrease (colour on a signed log scale). Instance-route share of the starting point in the titles.

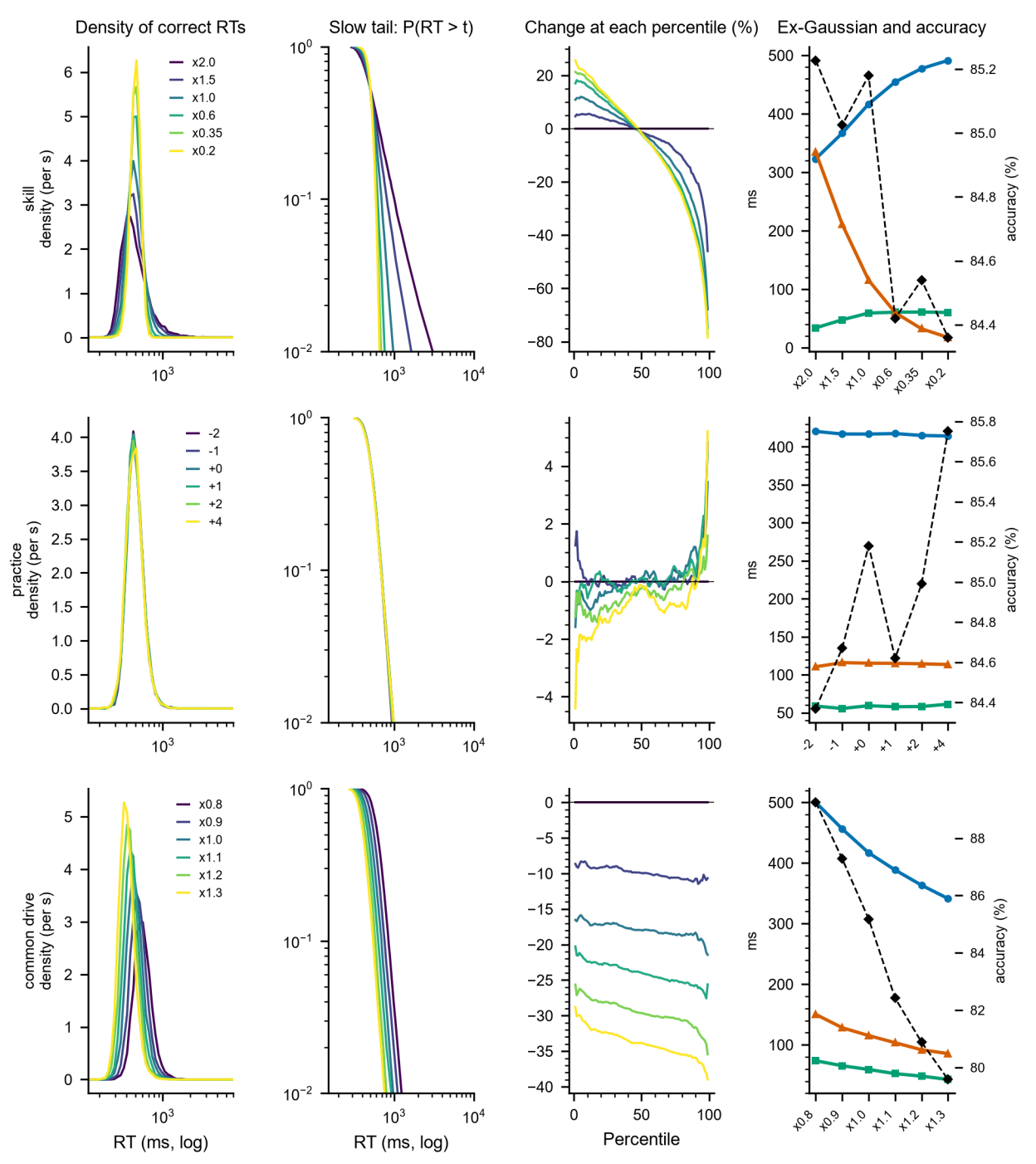



Figure 2. skill: drive spread u0_sd (lower = more skill); practice: stored instances logN; common drive u0 (u0_sd scaled with it). From the lexical-decision starting point (A2), each process alone at each setting (colours, light = highest). Columns: density of correct RTs (log time); survival function on log-log axes (the slow tail, to the 99th percentile); percent change at each percentile from the lowest setting; ex-Gaussian mu (blue circles), sigma (green squares) and tau (orange triangles) in ms, left axis, and accuracy (black diamonds, dashed), right axis, across settings.

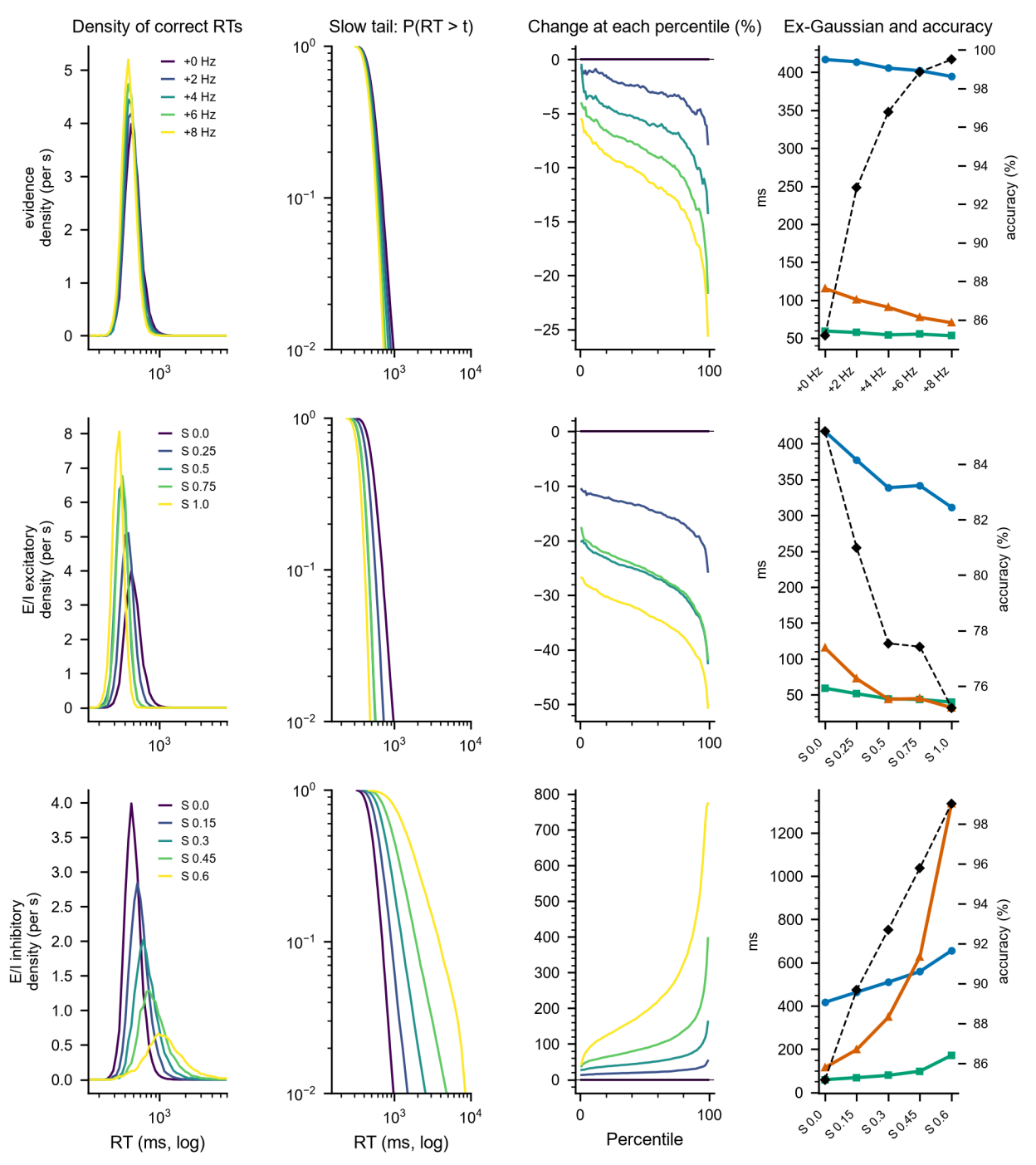



Figure 3. evidence: pool difference e; E/I control, toward excitation (R 1.05); E/I control, toward inhibition (R 1.3). From the lexical-decision starting point (A2), each process alone at each setting (colours, light = highest). Columns: density of correct RTs (log time); survival function on log-log axes (the slow tail, to the 99th percentile); percent change at each percentile from the lowest setting; ex-Gaussian mu (blue circles), sigma (green squares) and tau (orange triangles) in ms, left axis, and accuracy (black diamonds, dashed), right axis, across settings.

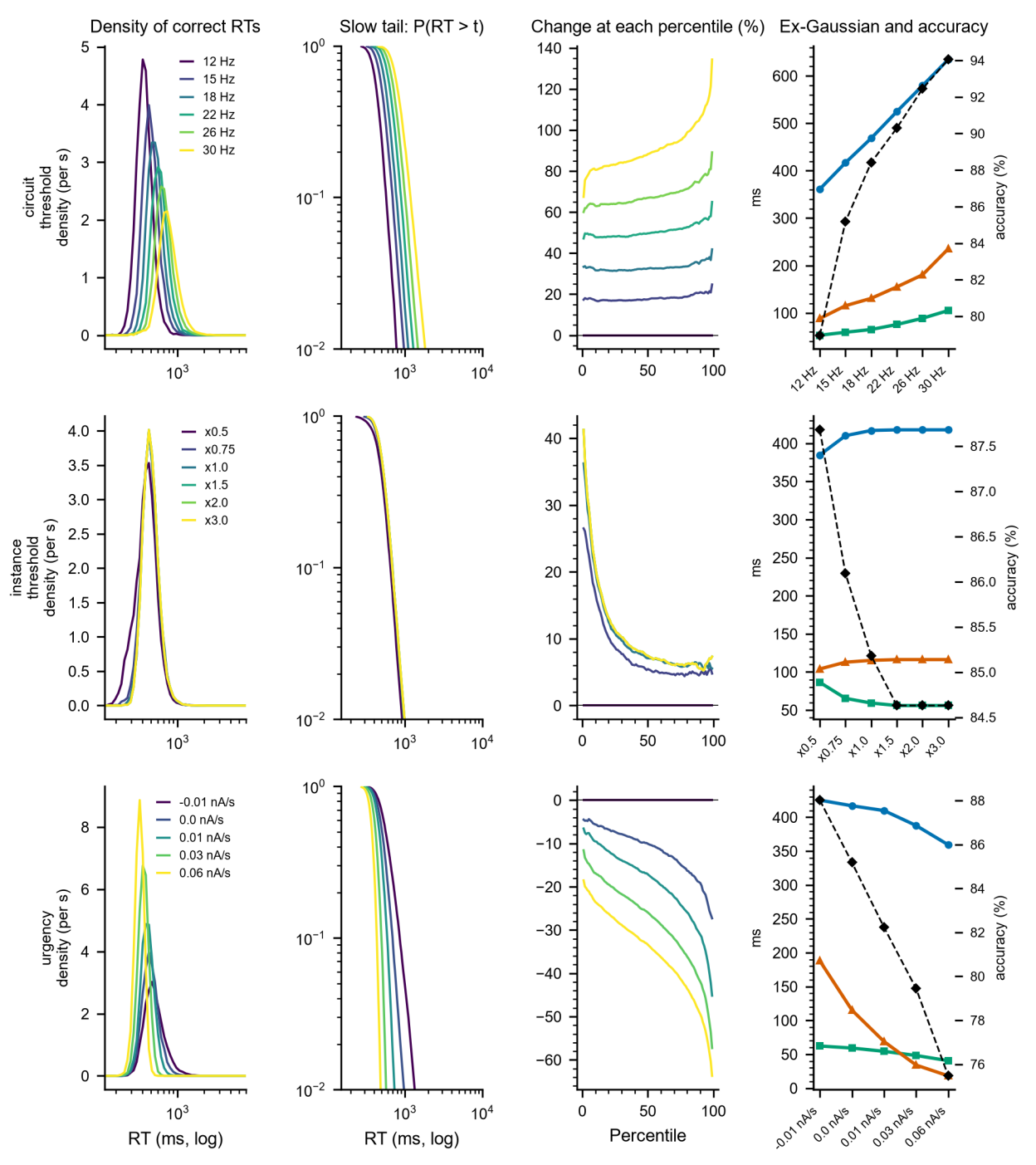



Figure 4. circuit threshold (commitment rate); instance-route threshold; urgency (input growing in the trial). From the lexical-decision starting point (A2), each process alone at each setting (colours, light = highest). Columns: density of correct RTs (log time); survival function on log-log axes (the slow tail, to the 99th percentile); percent change at each percentile from the lowest setting; ex-Gaussian mu (blue circles), sigma (green squares) and tau (orange triangles) in ms, left axis, and accuracy (black diamonds, dashed), right axis, across settings.

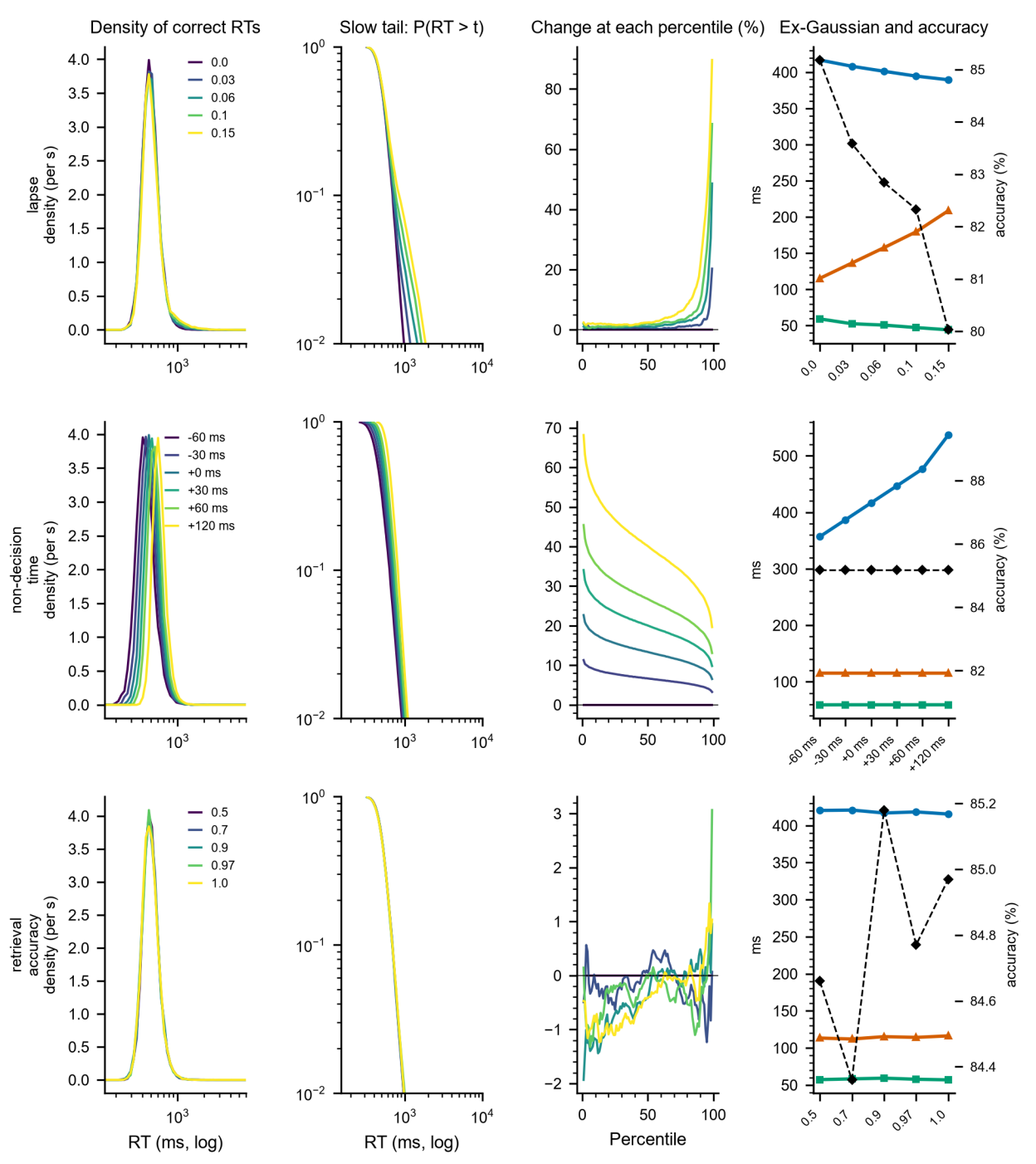



Figure 5. lapse (fraction of disengaged trials, gain 0.5); non-decision time ter_mu; retrieval accuracy (instance-route bias). From the lexical-decision starting point (A2), each process alone at each setting (colours, light = highest). Columns: density of correct RTs (log time); survival function on log-log axes (the slow tail, to the 99th percentile); percent change at each percentile from the lowest setting; ex-Gaussian mu (blue circles), sigma (green squares) and tau (orange triangles) in ms, left axis, and accuracy (black diamonds, dashed), right axis, across settings.

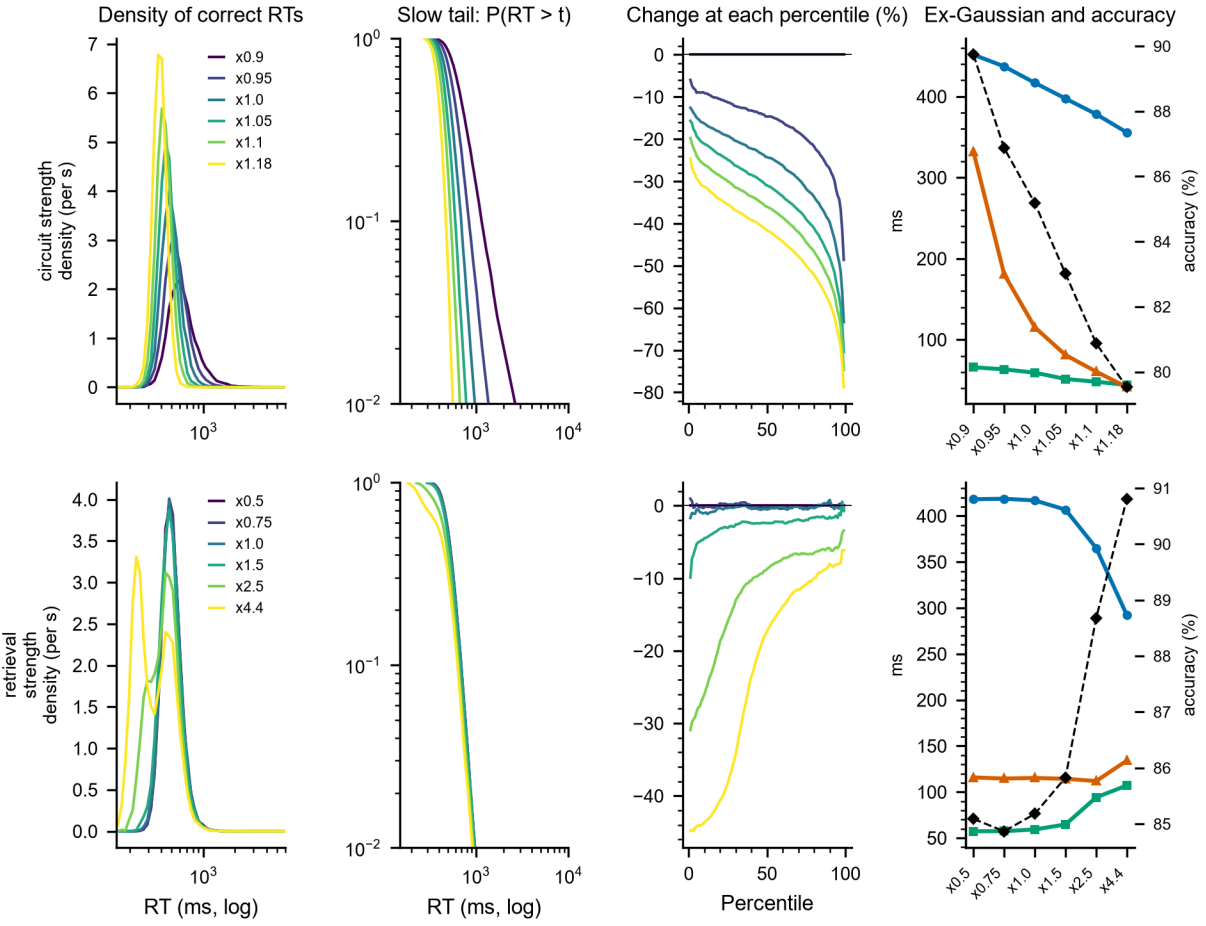



Figure 6. circuit strength (W_self, W_inhib scaled); retrieval strength (rho_max up, s_W down). From the lexical-decision starting point (A2), each process alone at each setting (colours, light = highest). Columns: density of correct RTs (log time); survival function on log-log axes (the slow tail, to the 99th percentile); percent change at each percentile from the lowest setting; ex-Gaussian mu (blue circles), sigma (green squares) and tau (orange triangles) in ms, left axis, and accuracy (black diamonds, dashed), right axis, across settings.

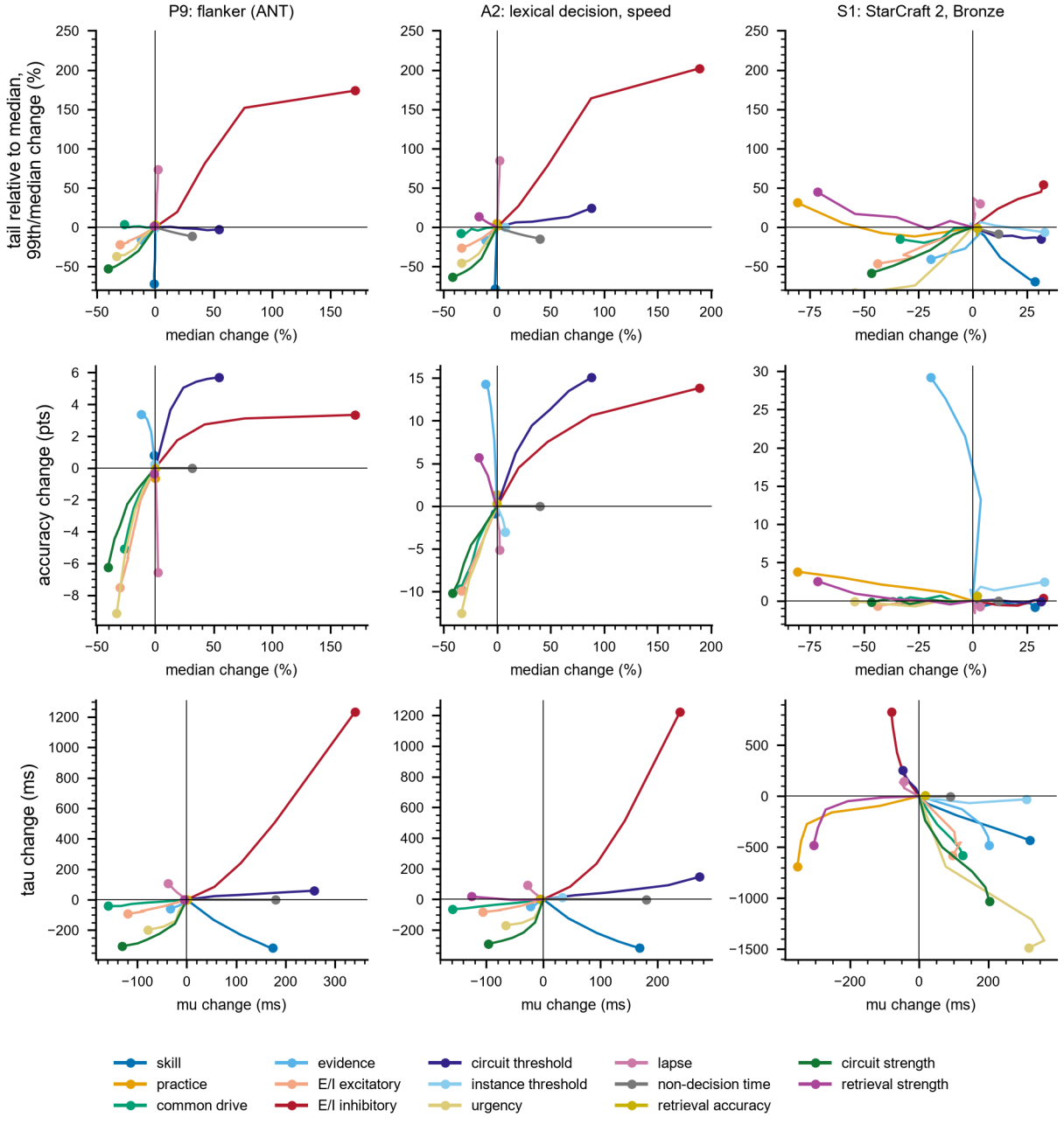



Figure 7. Shape space. Each line is one process traced from its lowest setting (origin) to its highest (dot), from each starting point. Top: change in the median against change in the tail relative to the median (99th/median), so a process that rescales the whole distribution runs along the horizontal axis and one that reshapes the tail runs vertically. Middle: speed against accuracy. Bottom: ex-Gaussian mu against tau. Top row clipped at -80% and +250%.

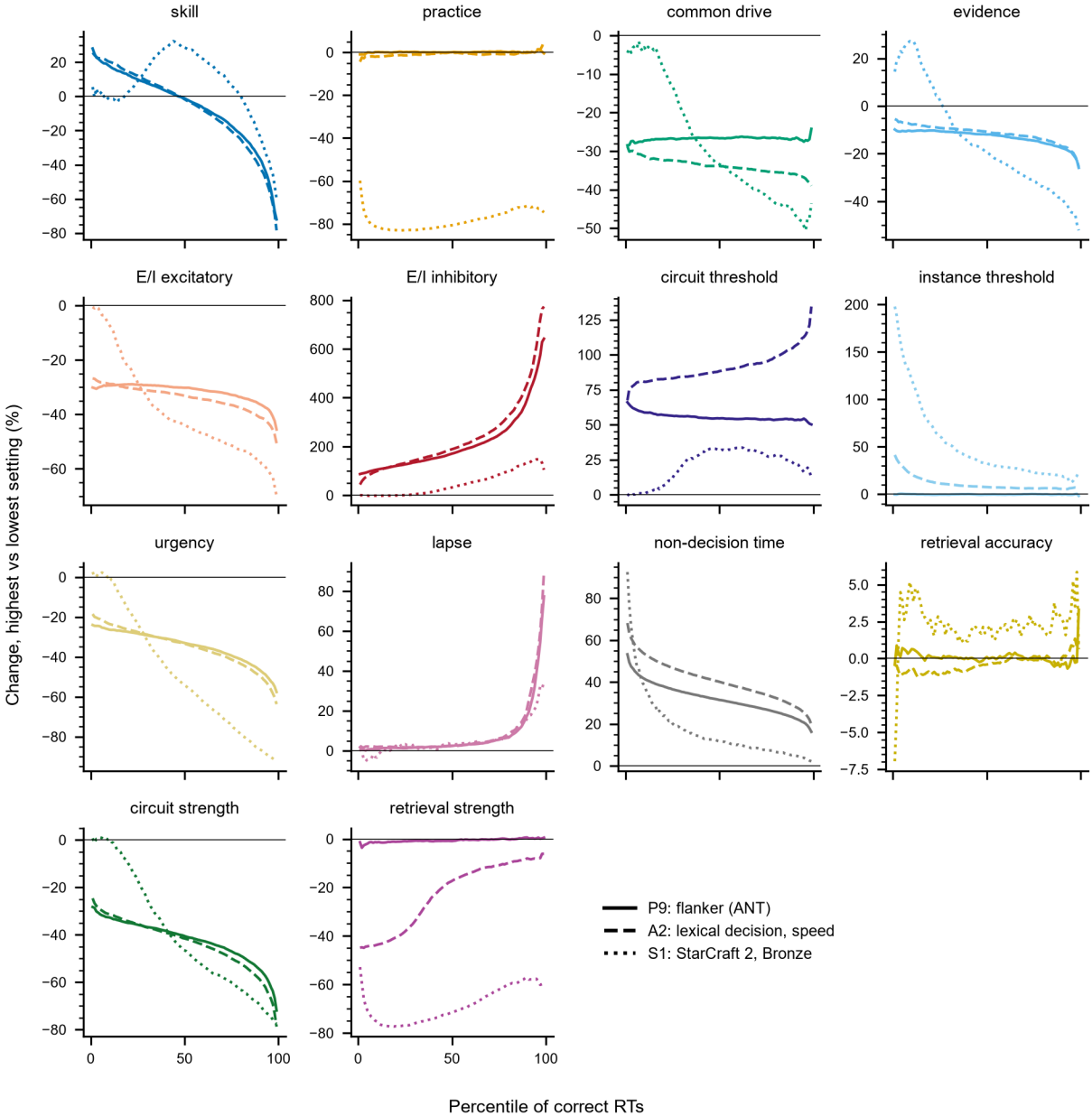



Figure 8. Where in the distribution each process acts: percent change at each percentile from its lowest to its highest setting, from the three starting points (line style). A flat line rescales or shifts the distribution; a rising line acts on the slow tail; a falling line acts on the fast edge. Each panel has its own y scale.

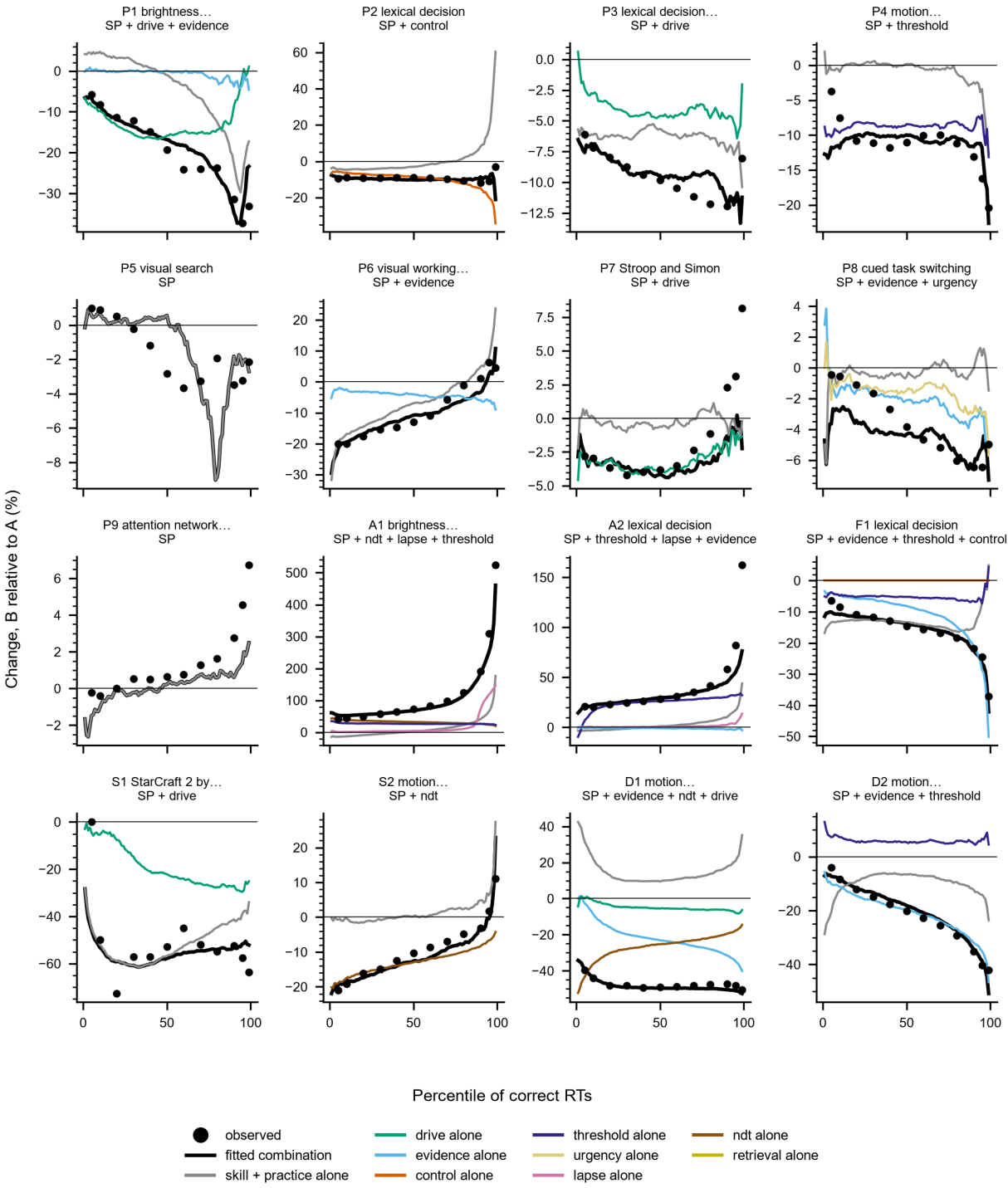



Figure 9. The fitted changes taken apart. For each contrast's chosen combination (stage 15; title), the observed change (points), the fitted combination (black), and each of its parts applied alone at its fitted value (coloured). Where the parts do not add up to the combination, the mechanisms interact.

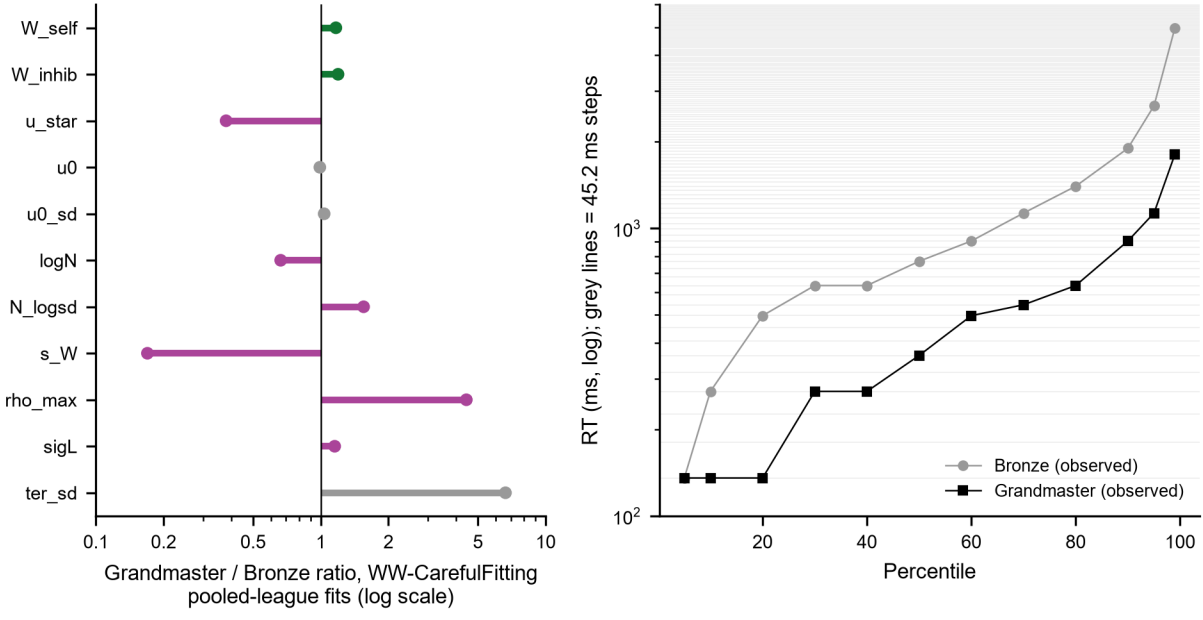



Figure 10. StarCraft 2, Bronze and Grandmaster. Left: ratio of each fitted parameter, Grandmaster to Bronze, in WW-CarefulFitting's pooled-league fits: circuit weights (green), retrieval route (purple; retrieval rate rho_max x4.4, drive gain set by u* 24.0 to 9.1 Hz, onset scale s_W x0.17), others (grey; ter_sd x6.6). Right: observed percentiles of S1, which lie on 45.2 ms recording steps (grey lines); Grandmaster's 5th, 10th and 20th percentiles are all 135.6 ms. Fits of S1 with these steps and with circuit and retrieval strength are in SkillPractice-StarCraftRefit_report.pdf.